

Personal Project Ideas

Minor AI for Society

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Introduction

For the personal project we have to include AI, which can be done in many different ways. At the beginning of the minor it was difficult to immediately determine which project would be the most suitable and realistic to execute within the available time.

During a feedback meeting with my teacher, Li Li (Fontys University of Applied Sciences), we discussed my current knowledge level and possible directions for a personal AI project. Based on this conversation, I decided that before committing to a specific project idea, it would be beneficial to strengthen my foundational understanding of machine learning.

For this reason, I started the Machine Learning Specialization by Andrew Ng. The goal of following this course is to establish a clearer baseline of what I already understand about AI and machine learning, as well as what skills I still need to develop. By building this foundation first, I will be better able to define a realistic and technically meaningful project later in the minor.

While I have not yet committed to one specific project, I have brainstormed several possible directions. These ideas help explore different applications of AI and allow for comparison in terms of feasibility, technical requirements, and learning potential. As I progress further in the course, I will refine these ideas and determine which one aligns best with my skills and interests.

The Ideas

As of now, I have three main project ideas that I am considering. Each of these projects incorporates AI in a different way and has its own unique focus and challenges. After completing the Machine Learning Specialization course, I will continue on working on these ideas, refining them, or maybe even coming up with new ones based on the knowledge I have gained. But for now, these three ideas serve as a starting point for exploration and comparison.

AI-Assisted Cinematic Tribute Editing Pipeline

This project extends a custom wrapper around `yt-dlp` into an AI-powered preprocessing tool for movie and TV tribute editing.

The system will automatically download selected videos and extract audio for analysis. Using speech-to-text and sentiment analysis, it will detect emotionally intense dialogue segments. Audio signal processing techniques will identify dramatic peaks and transitions.

A scoring engine will rank scenes based on emotional and cinematic weight. The tool will automatically generate timestamped highlights and export ready-to-edit clips. It will also extract impactful quotes that could be used for subtitle overlays.

All processing would run locally to ensure speed and privacy. The main goal is to reduce the amount of manual scrubbing required when creating tribute videos, while also enhancing the creative editing workflow.

This project would demonstrate applied AI techniques including natural language processing, audio signal processing, and automated media pipeline design.

Local AI-Powered Semantic File Finder

This project focuses on building a lightweight local AI tool that allows users to search their documents based on meaning instead of filenames.

The system would scan a user-selected folder and extract text from PDFs, Word documents, Markdown files, and TXT files. Documents would be split into smaller chunks (for example paragraph-level) and converted into embeddings using a small pre-trained transformer model.

These embeddings would be stored in a local vector database to enable efficient semantic search. Users could enter natural language queries to find relevant documents or passages.

The system would rank results based on semantic similarity and optionally combine this with keyword matching. A simple command line interface would display the file name, a short snippet of the relevant text, and a relevance score.

The goal is to create a practical AI-assisted document search tool that can quickly retrieve hard-to-find information. This project would demonstrate applications of AI in natural language processing, vector search, and lightweight system design.

AI-Powered Smart Desk Plant Monitor with Embedded Dashboard

This project explores the combination of embedded systems and AI by developing a smart desk plant monitoring system.

Each plant would be equipped with sensors that measure soil moisture, light intensity, temperature, and humidity. These sensors would report data to a central microcontroller that stores the readings in a local database.

A small touchscreen dashboard would provide an overview of all monitored plants, displaying color-coded health indicators. Selecting a specific plant would show detailed sensor readings, historical trends, and AI-generated care suggestions.

A TinyML model could be used to predict plant stress or watering needs based on patterns in the sensor data. Optionally, a small camera could allow AI-based plant classification to tailor care recommendations.

All processing would run locally without reliance on cloud services. The goal is to build a demonstrable and practical system that combines embedded hardware, sensor data analysis, and lightweight AI inference.

This project would demonstrate embedded system design, data collection, TinyML inference, and interactive interface development.

Next Steps

At this stage, these ideas serve as a starting point for exploration rather than finalized project proposals. As I continue progressing through the Machine Learning Specialization course, I will gain a clearer understanding of which techniques are realistic to implement and which project direction aligns best with my skills and learning goals.

Once a stronger technical baseline is established, I will refine one of these ideas into a concrete project plan with clearly defined scope, tools, and objectives.